



Task: SOC Fundamentals

1.SOC Purpose

A Security Operations Center (SOC) is responsible for continuous monitoring of networks, systems, and applications to detect and respond to cybersecurity threats.

The SOC team analyzes security alerts, investigates suspicious activities, and ensures timely incident response.

2.SOC Roles

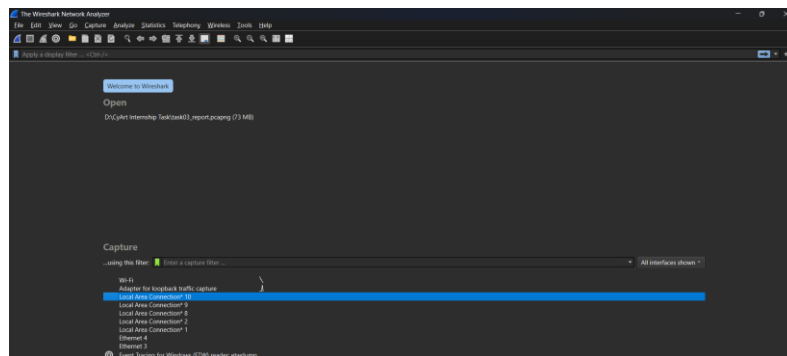
Role	Responsibility
Tier 1 Analyst	Monitor alerts and perform initial triage
Tier 2 Analyst	Investigate incidents and correlate logs
Tier 3 Analyst	Advanced threat hunting and malware analysis
SOC Manager	Oversees operations and response strategy

3.Security Monitoring Practice (Wireshark)

Step to Perform

→Open Wireshark:

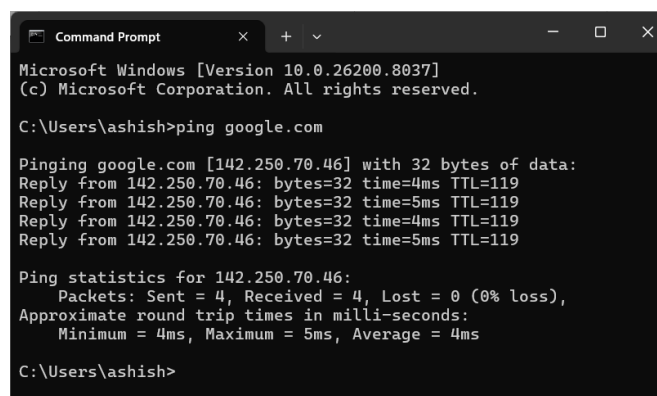
`sudo wireshark`



→Capture traffic on **eth0** or **lo**.

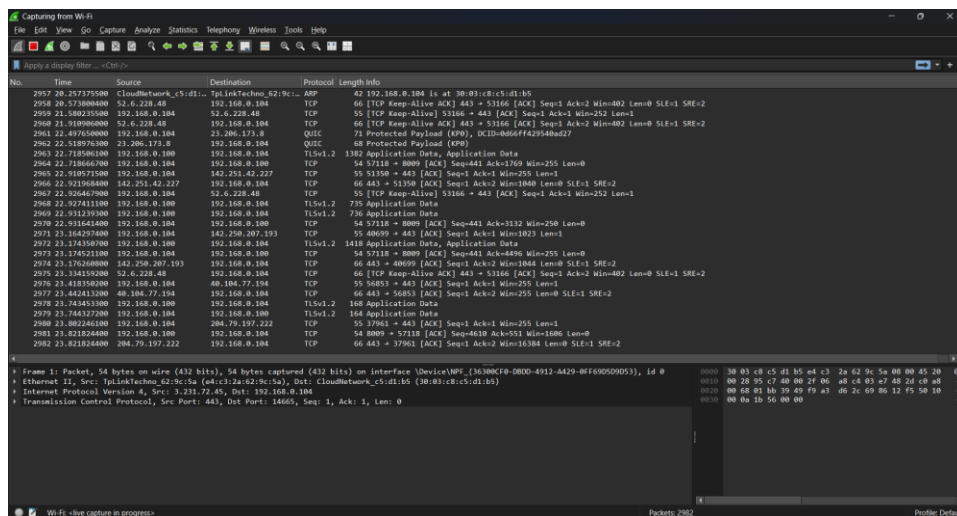
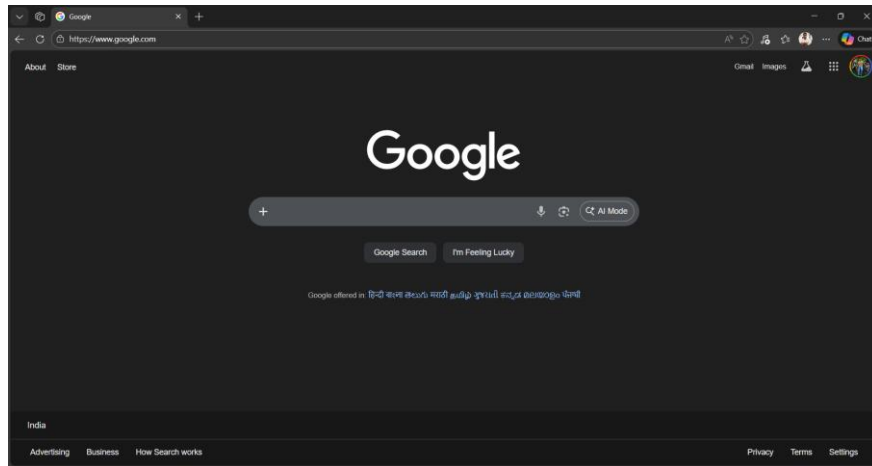
→Generate traffic:

`ping google.com`

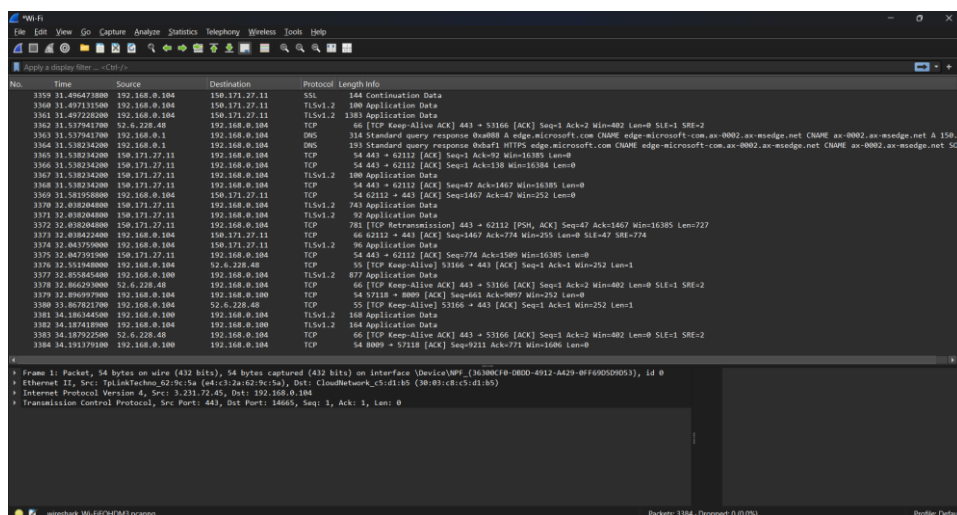




curl <https://google.com>



Start capture



Stop capture

1.Tcp

2.udp3.dns4.icmp



No.	Time	Source	Destination	Protocol	Length	Info
2334	11.962110300	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 2335)
2335	11.966281700	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=119 (request in 2334)
2742	12.589662300	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 2750)
2750	12.594757300	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=119 (request in 2742)
2925	14.085914400	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 2926)
2926	14.089775600	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=119 (request in 2925)
2934	15.494020300	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 2935)
2935	15.108025800	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=119 (request in 2934)

Frame 2750: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface Device\NPF_{36300CF8-DBD0-4912-A429-BFF69D509D53}, id 0

Ethernet II, Src: TpLinkTechno_62:9c:5a (e4:c3:2a:62:9c:5a), Dst: CloudNetwork_c5:d1:b5 (30:03:c8:c5:d1:b5)

Internet Protocol Version 4, Src: 142.250.70.46, Dst: 192.168.0.104

Internet Control Message Protocol

Screenshot

- DNS packets

No.	Time	Source	Destination	Protocol	Length	Info
18	6.072149000	192.168.0.104	192.168.0.1	DNS	72	Standard query 0x166b HTTPS www.bing.com
19	6.073073900	192.168.0.104	192.168.0.1	DNS	72	Standard query response 0x0a07 A www.bing.com
20	6.085847900	192.168.0.1	192.168.0.104	DNS	254	Standard query response 0x166b HTTPS www.bing.com CNAME www-ww.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e863
21	6.085847900	192.168.0.1	192.168.0.104	DNS	553	Standard query response 0x0a07 A www.bing.com CNAME www-ww.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e86303.d
633	7.799834900	192.168.0.104	192.168.0.1	DNS	74	Standard query 0xb0efc HTTPS www.google.com
634	7.800808200	192.168.0.104	192.168.0.1	DNS	74	Standard query 0xb0efc HTTPS www.google.com
635	7.804514500	192.168.0.1	192.168.0.104	DNS	347	Standard query response 0xb0efc HTTPS www.google.com HTTPS NS ns1.google.com NS ns2.google.com NS ns4.google.com NS ns3.google.com A 2
636	7.804514500	192.168.0.1	192.168.0.104	DNS	450	Standard query response 0xb0efc A www.google.com A 142.251.153.119 A 142.251.154.119 A 142.251.155.119 A 142.251.152.119 A 142.251.157
651	7.831779300	192.168.0.104	192.168.0.1	DNS	74	Standard query 0xf4a7 A www.google.com
656	7.865240900	192.168.0.1	192.168.0.104	DNS	450	Standard query response 0xf4a7 A www.google.com A 142.251.154.119 A 142.251.155.119 A 142.251.152.119 A 142.251.157.119 A 142.251.158
675	8.014785100	192.168.0.104	192.168.0.1	DNS	87	Standard query 0x582c HTTPS browser-events.data.msn.com
676	8.016103800	192.168.0.104	192.168.0.1	DNS	87	Standard query 0xf5e4 A browser.events.data.msn.com
681	8.017607800	192.168.0.1	192.168.0.104	DNS	269	Standard query response 0x582c HTTPS browser-events.data.msn.com CNAME global.asimov.events.data.trafficmanager.net CNAME onedcolprd
690	8.018981300	192.168.0.1	192.168.0.104	DNS	469	Standard query response 0xf5e4 A browser.events.data.msn.com CNAME global.asimov.events.data.trafficmanager.net CNAME onedcolprdeu8
781	8.319464600	192.168.0.104	192.168.0.1	DNS	85	Standard query 0x001a HTTPS lh3.googleusercontent.com
782	8.320545400	192.168.0.104	192.168.0.1	DNS	85	Standard query 0x062f A lh3.googleusercontent.com
783	8.323992200	192.168.0.104	192.168.0.1	DNS	77	Standard query 0x0a1f HTTPS fonts.gstatic.com
784	8.325236000	192.168.0.104	192.168.0.1	DNS	77	Standard query 0x0a26 A fonts.gstatic.com
871	8.330227800	192.168.0.1	192.168.0.104	DNS	171	Standard query response 0x001a HTTPS lh3.googleusercontent.com CNAME googlehosted.l.googleusercontent.com SOA ns1.google.com
880	8.331774600	192.168.0.1	192.168.0.104	DNS	357	Standard query response 0x0a1f HTTPS fonts.gstatic.com HTTPS NS ns4.google.com NS ns2.google.com NS ns3.google.com
969	8.342670300	192.168.0.1	192.168.0.104	DNS	385	Standard query response 0x062f A lh3.googleusercontent.com CNAME googlehosted.l.googleusercontent.com A 142.250.207.193 NS ns2.google
971	8.343799300	192.168.0.1	192.168.0.104	DNS	348	Standard query response 0x0a26 A fonts.gstatic.com A 142.250.71.99 NS ns4.google.com NS ns1.google.com NS ns3.google.com NS ns2.googl
1249	8.454519600	192.168.0.104	192.168.0.1	DNS	75	Standard query 0xb37e HTTPS www.gstatic.com
1252	8.455936100	192.168.0.104	192.168.0.1	DNS	75	Standard query 0x03ff A www.gstatic.com
1254	8.459346800	192.168.0.1	192.168.0.104	DNS	132	Standard query response 0xb37e HTTPS www.gstatic.com SOA ns1.google.com
1256	8.462376800	192.168.0.104	192.168.0.104	DNS	346	Standard query response 0x03ff A www.gstatic.com A 142.250.206.99 NS ns1.google.com NS ns3.google.com NS ns2.google.com NS ns4.google

Frame 18: Packet, 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface Device\NPF_{36300CF8-DBD0-4912-A429-BFF69D509D53}, id 0

Ethernet II, Src: TpLinkTechno_c5:d1:b5 (30:03:c8:c5:d1:b5), Dst: TpLinkTechno_62:9c:5a (e4:c3:2a:62:9c:5a)

Internet Protocol Version 4, Src: 192.168.0.104, Dst: 192.168.0.1

User Datagram Protocol, Src Port: 49445, Dst Port: 53

Domain Name System (query)

- TCP traffic

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	3.231.72.45	192.168.0.104	TCP	54	443 → 14665 [ACK] Seq=1 Ack=1 Win=0 Len=0
2	0.000126600	192.168.0.104	3.231.72.45	TCP	54	[TCP ACKed unseen segment] 14665 → 443 [ACK] Seq=1 Ack=2 Win=255 Len=0
3	0.078318100	192.168.0.104	52.6.228.48	TCP	55	53166 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
4	0.4097970200	52.6.228.48	192.168.0.104	TCP	66	443 → 53166 [ACK] Seq=1 Ack=2 Win=402 Len=0 SLE=1 SRE=2
5	1.5093636700	192.168.0.104	52.6.228.48	TCP	55	[TCP Keep-Alive] 53166 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6	1.833793400	52.6.228.48	192.168.0.104	TCP	66	[TCP Keep-Alive ACK] 443 → 53166 [ACK] Seq=1 Ack=2 Win=402 Len=0 SLE=1 SRE=2
7	2.201160200	192.168.0.104	150.171.28.11	TCP	55	5865 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
8	2.2216002500	150.171.28.11	192.168.0.104	TCP	66	443 → 5865 [ACK] Seq=1 Ack=2 Win=16385 Len=0 SLE=1 SRE=2
9	2.8048666900	192.168.0.104	192.168.0.100	TLSv1.2	164	Application Data
10	2.8321545900	192.168.0.100	192.168.0.104	TLSv1.2	164	Application Data
11	2.836453700	192.168.0.104	52.6.228.48	TCP	55	[TCP Keep-Alive] 53166 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
12	2.8845363600	192.168.0.104	192.168.0.100	TCP	54	57118 → 8009 [ACK] Seq=111 Ack=111 Win=255 Len=0
13	1.1571213500	52.6.228.48	192.168.0.104	TCP	66	[TCP Keep-Alive ACK] 443 → 53166 [ACK] Seq=1 Ack=2 Win=402 Len=0 SLE=1 SRE=2
14	1.159186400	192.168.0.104	52.6.228.48	TCP	55	[TCP Keep-Alive] 53166 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
15	4.4898395900	52.6.228.48	192.168.0.104	TCP	66	[TCP Keep-Alive ACK] 443 → 53166 [ACK] Seq=1 Ack=2 Win=402 Len=0 SLE=1 SRE=2
16	5.507427800	192.168.0.104	52.6.228.48	TCP	55	[TCP Keep-Alive] 53166 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
17	6.0025037600	52.6.228.48	192.168.0.104	TCP	66	[TCP Keep-Alive ACK] 443 → 53166 [ACK] Seq=1 Ack=2 Win=402 Len=0 SLE=1 SRE=2
24	6.174586200	192.168.0.104	172.188.155.25	TCP	55	5835 → 443 [ACK] Seq=1 Ack=1 Win=250 Len=1
83	6.236810700	172.188.155.25	192.168.0.104	TCP	66	443 → 5835 [ACK] Seq=1 Ack=2 Win=274 Len=0 SLE=1 SRE=2
625	7.037914800	192.168.0.104	52.6.228.48	TCP	55	[TCP Keep-Alive] 53166 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
628	7.244577300	52.6.228.48	192.168.0.104	TCP	66	[TCP Keep-Alive ACK] 443 → 53166 [ACK] Seq=1 Ack=2 Win=402 Len=0 SLE=1 SRE=2
650	7.836495400	192.168.0.104	192.168.0.100	TLSv1.2	164	Application Data
657	7.865240900	192.168.0.100	192.168.0.104	TLSv1.2	164	Application Data
662	7.908407800	192.168.0.104	192.168.0.100	TCP	54	57118 → 8009 [ACK] Seq=221 Ack=221 Win=255 Len=0
677	8.017439600	192.168.0.104	20.42.65.84	TCP	1494	15418 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1440 [TCP PDU reassembled in 680]
678	8.017439600	192.168.0.104	20.42.65.84	TCP	1494	15418 → 443 [ACK] Seq=1441 Ack=1 Win=255 Len=1440 [TCP PDU reassembled in 680]
679	8.017439600	192.168.0.104	20.42.65.84	TCP	1494	15418 → 443 [ACK] Seq=3881 Ack=1 Win=255 Len=1440 [TCP PDU reassembled in 680]

Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface Device\NPF_{36300CF8-DBD0-4912-A429-BFF69D509D53}, id 0

Ethernet II, Src: TpLinkTechno_62:9c:5a (e4:c3:2a:62:9c:5a), Dst: CloudNetwork_c5:d1:b5 (30:03:c8:c5:d1:b5)

Internet Protocol Version 4, Src: 3.231.72.45, Dst: 192.168.0.104

Transmission Control Protocol, Src Port: 443, Dst Port: 14665, Seq: 1, Ack: 1, Len: 0

- ICMP traffic



No.	Time	Source	Destination	Protocol	Length	Info
2334	11.962118300	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in 2335)
2335	11.9660281700	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=5/1280, ttl=119 (request in 2334)
2742	12.989662300	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=6/1536, ttl=128 (reply in 2750)
2750	12.994734300	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=6/1536, ttl=119 (request in 2742)
2925	14.085914400	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 2926)
2926	14.089775600	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=119 (request in 2925)
2934	15.094624300	192.168.0.104	142.250.70.46	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 2935)
2935	15.100025800	142.250.70.46	192.168.0.104	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=119 (request in 2934)

Frame 2750: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{36300CF0-0B0D-4912-A429-0FF6905D9D53}, id 0
Ethernet II, Src: TpLinkTechno_62:9c:5a (e4:c3:2a:62:9c:5a), Dst: CloudNetwork_c5:d1:b5 (30:03:c8:c5:d1:b5)
Internet Protocol Version 4, Src: 142.250.70.46, Dst: 192.168.0.104
Internet Control Message Protocol

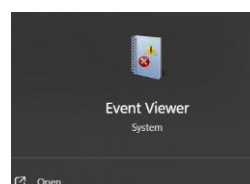
Security Monitoring

Network traffic was captured using Wireshark to analyze communication patterns. Protocols such as TCP, UDP, DNS, and ICMP were identified during the capture. Traffic analysis helps detect anomalies, suspicious activity, and unauthorized communication within the network.

4.Log Analysis Practice

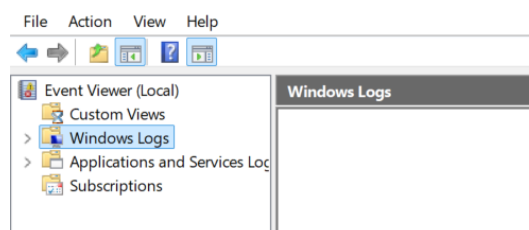
If you have Windows VM

→Open Event Viewer

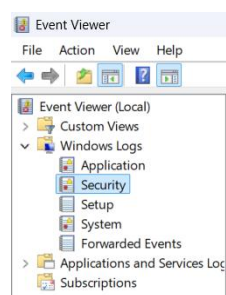


→Navigate:

Windows Logs



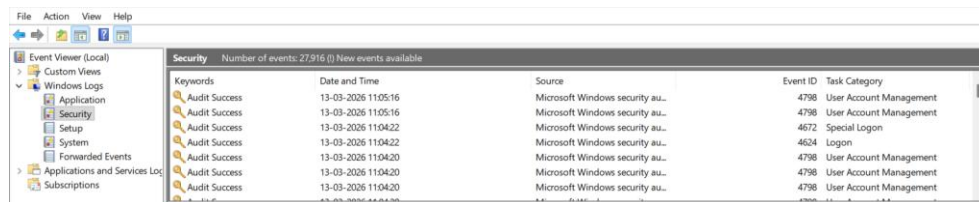
→ Security





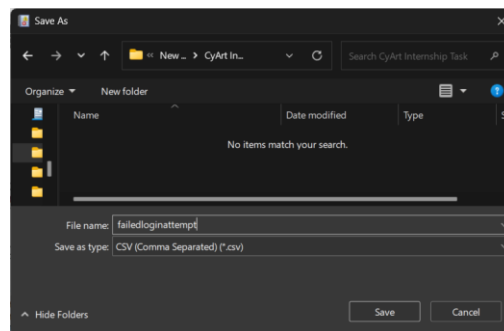
→Filter for Event ID:

4625,4624



→This shows **failed login attempts**.

Export results to CSV.



If you only have Kali Linux

```
(abhishek@kali)-[~]  
$ ssh test@localhost  
test@localhost's password:
```

```
(abhishek@kali)-[~]  
$ ssh test@localhost  
test@localhost's password:  
Permission denied, please try again.  
test@localhost's password:  
Permission denied, please try again.  
test@localhost's password:  
test@localhost: Permission denied (publickey,password).
```

→Use authentication logs:

```
sudo cat /var/log/auth.log
```

→Search failed logins:

```
grep "Failed password" /var/log/auth.log
```

```
sudo journalctl -u ssh
```

Screenshot

- Failed login entries



```
(abhishek@kali)-[~]
$ sudo journalctl -u ssh
Sep 23 20:36:51 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Sep 23 20:36:52 kali sshd[2776]: Server listening on 0.0.0.0 port 22.
Sep 23 20:36:52 kali sshd[2776]: Server listening on :: port 22.
Sep 23 20:36:52 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
-- Boot 876a02478d5f45f6b019fad7423577f9 --
Feb 26 15:49:56 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Feb 26 15:49:56 kali sshd[2214]: Server listening on 0.0.0.0 port 22.
Feb 26 15:49:56 kali sshd[2214]: Server listening on :: port 22.
Feb 26 15:49:56 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 26 15:50:50 kali sshd-session[2772]: Invalid user test from ::1 port 48002
Feb 26 15:51:03 kali sshd-session[2772]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:51:03 kali sshd-session[2772]: pam_unix(sshd:auth): authentication failure; logname= uid=0
Feb 26 15:51:03 kali sshd-session[2772]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:51:03 kali sshd-session[2772]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:51:05 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 s
Feb 26 15:51:09 kali sshd-session[2772]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:51:09 kali sshd-session[2772]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:51:09 kali sshd-session[2772]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:51:11 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 s
Feb 26 15:51:18 kali sshd-session[2772]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:51:18 kali sshd-session[2772]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:51:18 kali sshd-session[2772]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:51:21 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 s
Feb 26 15:51:22 kali sshd-session[2772]: Connection closed by invalid user test ::1 port 48002 [prea
Feb 26 15:51:22 kali sshd-session[2772]: PAM 2 more authentication failures; logname= uid=0 euid=0 t
Feb 26 15:54:56 kali sshd-session[4861]: Invalid user test from ::1 port 42278
Feb 26 15:55:08 kali sshd-session[4861]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:55:08 kali sshd-session[4861]: pam_unix(sshd:auth): authentication failure; logname= uid=0
Feb 26 15:55:08 kali sshd-session[4861]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:55:08 kali sshd-session[4861]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:55:10 kali sshd-session[4861]: Failed password for invalid user test from ::1 port 42278 s
Diner 1-31 skipping
```

```
Sep 23 20:36:51 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Sep 23 20:36:52 kali sshd[2776]: Server listening on 0.0.0.0 port 22.
Sep 23 20:36:52 kali sshd[2776]: Server listening on :: port 22.
Sep 23 20:36:52 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
-- Boot 876a02478d5f45f6b019fad7423577f9 --
Feb 26 15:49:56 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Feb 26 15:49:56 kali sshd[2214]: Server listening on 0.0.0.0 port 22.
Feb 26 15:49:56 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 26 15:50:50 kali sshd-session[2772]: Invalid user test from ::1 port 48002
Feb 26 15:51:03 kali sshd-session[2772]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:51:03 kali sshd-session[2772]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost:::1
Feb 26 15:51:03 kali sshd-session[2772]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:51:03 kali sshd-session[2772]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:51:05 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 ssh2
Feb 26 15:51:09 kali sshd-session[2772]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:51:09 kali sshd-session[2772]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:51:09 kali sshd-session[2772]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:51:11 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 ssh2
Feb 26 15:51:18 kali sshd-session[2772]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:51:18 kali sshd-session[2772]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:51:18 kali sshd-session[2772]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:51:21 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 ssh2
Feb 26 15:51:22 kali sshd-session[2772]: Connection closed by invalid user test ::1 port 48002 [preauth]
Feb 26 15:51:22 kali sshd-session[2772]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost:::1
Feb 26 15:54:56 kali sshd-session[4861]: Invalid user test from ::1 port 42278
Feb 26 15:55:08 kali sshd-session[4861]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:55:08 kali sshd-session[4861]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost:::1
Feb 26 15:55:08 kali sshd-session[4861]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:55:08 kali sshd-session[4861]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:55:10 kali sshd-session[4861]: Failed password for invalid user test from ::1 port 42278 ssh2
Feb 26 15:55:18 kali sshd-session[4861]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:55:18 kali sshd-session[4861]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:55:18 kali sshd-session[4861]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:55:20 kali sshd-session[4861]: Failed password for invalid user test from ::1 port 42278 ssh2
Feb 26 15:55:27 kali sshd-session[4861]: pam_unix(sshd:auth): check pass; user unknown
Feb 26 15:55:27 kali sshd-session[4861]: pam_winbind(sshd:auth): getting password (0x00000388)
Feb 26 15:55:27 kali sshd-session[4861]: pam_winbind(sshd:auth): pam_get_item returned a password
Feb 26 15:55:29 kali sshd-session[4861]: Failed password for invalid user test from ::1 port 42278 ssh2
Feb 26 15:55:29 kali sshd-session[4861]: Connection closed by invalid user test ::1 port 42278 [preauth]
Feb 26 15:55:29 kali sshd-session[4861]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost:::1
```

Log Analysis

System authentication logs were analyzed to identify failed login attempts.

These logs help detect brute-force attacks and unauthorized access attempts. Log analysis is a critical SOC function used for early threat detection.

5.Configure Alert Rule Simulation

You will simulate **multiple failed SSH login attempts**.

Generate Failed Logins

→ssh test@localhost

Enter wrong password multiple times.

→Now check logs:

grep "Failed password" /var/log/auth.log



```
sudo journalctl -u ssh | grep "Failed"
```

```
(abhishek@kali)~$ sudo journalctl -u ssh | grep "Failed"
Feb 26 15:51:05 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 ssh2
Feb 26 15:51:11 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 ssh2
Feb 26 15:51:21 kali sshd-session[2772]: Failed password for invalid user test from ::1 port 48002 ssh2
Feb 26 15:55:10 kali sshd-session[4861]: Failed password for invalid user test from ::1 port 42278 ssh2
Feb 26 15:55:20 kali sshd-session[4861]: Failed password for invalid user test from ::1 port 42278 ssh2
Feb 26 15:55:29 kali sshd-session[4861]: Failed password for invalid user test from ::1 port 42278 ssh2
Feb 26 16:05:41 kali sshd-session[10225]: Failed password for invalid user test from ::1 port 44556 ssh2
Feb 26 16:05:48 kali sshd-session[10225]: Failed password for invalid user test from ::1 port 44556 ssh2
Feb 26 16:05:53 kali sshd-session[10225]: Failed password for invalid user test from ::1 port 44556 ssh2
Feb 26 16:06:16 kali sshd-session[10518]: Failed password for invalid user admin from ::1 port 36118 ssh2
Feb 26 16:06:23 kali sshd-session[10518]: Failed password for invalid user admin from ::1 port 36118 ssh2
Feb 26 16:06:27 kali sshd-session[10518]: Failed password for invalid user admin from ::1 port 36118 ssh2
Feb 26 16:09:16 kali sshd-session[11985]: Failed password for invalid user test from ::1 port 35120 ssh2
Feb 26 16:09:22 kali sshd-session[11985]: Failed password for invalid user test from ::1 port 35120 ssh2
Feb 26 16:09:28 kali sshd-session[11985]: Failed password for invalid user test from ::1 port 35120 ssh2
Mar 13 11:51:09 kali sshd-session[5511]: Failed password for invalid user test from ::1 port 55214 ssh2
Mar 13 11:51:20 kali sshd-session[5511]: Failed password for invalid user test from ::1 port 55214 ssh2
Mar 13 11:51:27 kali sshd-session[5511]: Failed password for invalid user test from ::1 port 55214 ssh2
Mar 13 12:01:45 kali sshd-session[10758]: Failed password for invalid user test from ::1 port 37050 ssh2
Mar 13 12:01:53 kali sshd-session[10758]: Failed password for invalid user test from ::1 port 37050 ssh2
Mar 13 12:02:00 kali sshd-session[10758]: Failed password for invalid user test from ::1 port 37050 ssh2
```

You should see multiple entries.

Alert Rule Simulation

Multiple failed SSH login attempts were generated to simulate a brute-force attack.

Authentication logs recorded repeated failed login events, demonstrating how SOC systems detect suspicious login patterns.

6.Incident Documentation

Date	Source IP	Event	Description	Action Taken
12-03-2026	127.0.0.1	Failed SSH Login	Multiple failed SSH login attempts detected	Logs analyzed and suspicious activity investigated
12-03-2026	127.0.0.1	Failed SSH Login	Repeated authentication failures from the same source IP	Monitored system logs and verified potential brute-force behavior

Incident Report Example

During monitoring, repeated failed SSH login attempts were detected from the same source IP address. This pattern indicates a potential brute-force attack. The incident was analyzed through system logs and network traffic monitoring.

Indicators Observed

- Multiple failed SSH authentication attempts
- Repeated access attempts from the same source IP
- Target service: **SSH (Port 22)**

Investigation Tools Used

- System authentication logs (journalctl)
- Network traffic monitoring (Wireshark)



7. Threat Hunting

Threat hunting involves proactively searching for suspicious activities in system logs and network traffic to identify potential security threats.

Steps Performed

Authentication logs were analyzed using the following commands:

```
sudo journalctl -u ssh  
sudo journalctl -u ssh | grep "Failed"
```

The logs were reviewed to identify:

- Repeated failed login attempts
- Multiple authentication failures from the same source IP
- Unusual login activity

Findings

Repeated failed SSH login attempts from the same source IP address were detected. This behavior is commonly associated with **brute-force login attempts**.

Proactive log analysis and threat hunting techniques help SOC analysts detect potential attacks early and prevent further compromise.

8. Tools Overview

Tool	Purpose
Wireshark	Network traffic analysis
SIEM (Elastic/Splunk)	Log monitoring
Snort	Intrusion detection
Nessus	Vulnerability scanning

9. Security Concepts

CIA Triad

The **CIA Triad** is a fundamental model in cybersecurity that represents the three core principles required to protect information systems and data.

Confidentiality

It ensures that sensitive information is accessible only to authorized users and protected from unauthorized access or disclosure.



Techniques such as encryption, authentication, and access controls are commonly used to maintain confidentiality.

Integrity

It ensures that data remains accurate, consistent, and unaltered during storage, processing, or transmission.

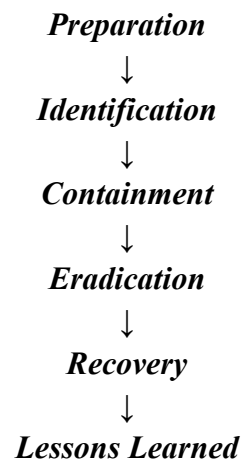
Mechanisms such as hashing, checksums, and digital signatures help detect and prevent unauthorized modifications.

Availability

It ensures that systems, services, and data are accessible to authorized users whenever they are needed.

This is achieved through system redundancy, regular maintenance, backups, and protection against attacks such as Denial-of-Service (DoS).

10. Incident Response Lifecycle



The **Incident Response Lifecycle** describes the steps security teams follow to detect, manage, and recover from a security incident.

1. Preparation

Preparation means getting ready in advance to handle possible security incidents.

- Security teams set up monitoring tools, policies, and incident response plans.
- Systems are regularly updated, backups are maintained, and staff are trained on how to respond to incidents.

2. Identification

In this stage, the security team detects that a security incident may have occurred.

- Analysts monitor alerts, logs, and unusual system activity.
- They analyze the data to confirm whether it is a real attack or a false alert.



3. Containment

Containment focuses on stopping the attack from spreading to other systems.

- The affected system may be isolated from the network.
- Suspicious IP addresses, accounts, or processes may be blocked.

4. Eradication

This step removes the root cause of the attack from the system.

- Malware, malicious files, or unauthorized access are removed.
- Vulnerabilities are patched and system security is improved.

5. Recovery

Recovery means restoring systems and services back to normal operation.

- Clean backups may be restored if needed.
- Systems are monitored carefully to make sure the attack does not return.

6. Lessons Learned

After the incident is handled, the team reviews what happened.

- The incident response process is analyzed to identify improvements.
- Security policies and defenses are updated to prevent similar attacks in the future.

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